

Project Report

Vietnamese Aspect-based Sentiment Analysis

Course: IT4772E - Natural Language Processing

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Abstract

In this report, we conducted our study on Aspect-based Sentiment Analysis, which generates the sentiments from human opinion in different aspects. The dataset that we use for this task is from VLSP 2018 - Evaluation Campaign. This data contains reviews that assess aspects in the domain of restaurant, with three polarities: positive, negative or neural. To test the effectiveness of both traditional and modern learning methods , we propose different approaches: Support vector machine (SVM), CNN, RoBERTa and BART. The results show that RoBERTa approach method has reached the state-of-the-art on Vietnamese ABSA task with this dataset. Github: `https://github.com/minhngt62/nlp-vabsa`.

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1 Introduction

Understanding the comments, feedback, and opinions of users, clients, and customers has always been crucial in all fields of life. Unfortunately, manually handling the vast amount of textual content is now unachievable. This problem leads to the emergence of the research field sentiment analysis (SA), which helps automatically analyze the opinions of users. With an increase in the complexity of products, companies also need to recognize comments on smaller aspects of the products; therefore, the SA system has to analyze more detailed parts of users' feedback to recognize more fine-grained aspect-level opinions and sentiments. This field of study is named as Aspect-Based Sentiment Analysis (ABSA), which includes many a variety of helping tasks.

In this project for the course Introduction to Natural Language Processing, our group will solve the ABSA problem on a Vietnamese ABSA-VLSP 2018 dataset [2] from the VLSP competition. This dataset requires solutions for the Aspect Category Sentiment Analysis (ACSA) problem, a compound ABSA problem including two tasks: Aspect Category Detection (ACD) and Aspect Sentiment Classification (ASC). ACD asks a system to retrieve all mentioned aspects from an opinion, and ASC requires classifying the sentiment related to each aspect. More detailed information about the dataset and the problem is presented in the later part. As the competition was held 5 years ago (in 2018), the performances of the three participating teams were not promising. With the appearance of many good techniques in Natural Language Processing recently, we want to apply them to solve this problem and compare the performance with the traditional methods.

This progress report includes three parts. This part (introduction) brings a quick review of our problem. The second part will introduce some methods that we intend to apply in this project. The final part present detail information about the dataset, metrics, and experiments.

2 Related works

Aspect category sentiment analysis (ACSA) [9] seeks to detect the mentioned aspect categories as well as the emotion polarity associated with them in a comment. Due to its twofold output, this task is widely used in industries. With the knowledge gained after courses at university, we want to apply and optimize some of them in this task with a Vietnamese dataset for ACSA - the VLSP ABSA dataset.

ACSA can be split into two small tasks of Aspect Category Detection (ACD) and Aspect Sentiment Classification (ASC) or can be combined as one ASC task with one additional label, which means that the tag (category) is not mentioned. Therefore, the most famous strategy is to solve the task as classification problems. For the two-task method, the ACD task is a multi-label classification problem (treating each category as a label), and the ASC task is a multi-class classification problem (where each sentiment polarity is a class) for each detected aspect category. Each of our first two models follows one of the two strategies mentioned above, using simple techniques like Support Vector Machine (SVM), Random Forest (RF) and Gated Convolution Neural Networks (taken from [15]) (Gated CNN). The utilization of Bags of Words (BoW) or CNN helps the two models can capture the semantic knowledge in a part of the context, to detect categories and classify polarity. With the existence of pre-trained Transformers [10] based models, especially encoder-only models like BERT [11], RoBERTa [4], and their variants, which can encode all the information of a long text efficiently, we now can solve ACSA as the prob-

lem of text classification with the input combining the comment and a category. Our third model is based on this idea with the application of a RoBERTa version, which is only trained on Vietnamese corpus, PhoBERT [3]. Liu et al. [7] recently used the Sequence-to-sequence (Seq2Seq) modeling paradigm to the ACSA problem. They employ natural language words to express the intended output and get highly promising outcomes based on a pre-trained generative model. Our last model will follow this strategy to be one of the earliest efforts to construct a generative model for Vietnamese ABSA. We use BARTpho [13] for segmented words as the Seq2Seq architecture.

3 Methodology

3.1 LDA and Machine Learning

Following the section 4.1.2, the ongoing problem is multitask classification in which reviews belong to one or more aspect categories and only one polarity. For the traditional machine learning models to approach such a task comfortably, we transform multitask problem into multi-label classification in which an aspect is encoded using label encoding, which ranges from 0 to 3. The values from 1 to 3 represent the polarities of the corresponding aspects, and zero (i.e., "0") does mean an aspect does not relate to the review.

Text processing Discussing about stopword removal, we think that it had better not be included here since our topic - sentiment analysis is quite more sensitive to such words that often contain reviewers' attitudes (e.g., hate, satisfied,...) rather than document classification. For example, "I told you that she was not happy" remains ["told", "happy"] that completely changes the overall meaning of the sentence.

Text vectorization For the machine learning methods such as Support Vector Machine (SVM) to understand natural languages, text vectorization is of importance. Thus, we apply Bag of Words (BoW) on the n-gram vocabulary (i.e., $n = 2$) of the corpus so that the context can be exploited as well, even just slightly. Since the number of dimension is huge, we leverage Latent Dirichlet Allocation (LDA) to map the vectorized space of text into the space each of whose axes represents for a couple of aspect and polarity (i.e., a topic). From our re-definition for the problem, each of aspect owns four classes (i.e., 3 polarity and 1 not-belong) and there are 12 aspects, which means 48 topics/axes after LDA estimation. This allows the traditional machine learning algorithms such as support vector machine to be trained in affordable time and resource. One more detail is that TF-IDF has been used once after BoW; however, it yields no performance boost while slows down the training process. Therefore, we keep BoW solely as the input for LDA.

Modelling We approach the problem with two representatives which are support vector machine (SVM) and random forest (RF). The former has such a long development and proofs with natural language processing while the latter is able to be trained parallelly with well-known extraordinary performance, compared to other boosting algorithms.

To attain an appropriate set of parameters, we wrap all the processing steps above with the models as a whole completed pipeline. Randomized search (i.e., 100 iterations) is then applied to the pipelines to find the potential combinations, under cross-validation. The cross-validation is set to 5 folds, fitting well with the resources we have; and both training and dev set is included to construct the folds. After that, we test the best pipelines on the test set

to extract the best of best under three metrics defined later.

3.2 Gated CNN

In this section, we present a simple and efficient method, namely Gated Convolutional network [15]. It is built on convolutional layers and gating units. Each convolutional filter computes n-gram features at different granularities from the embedding vectors at each position individually. The gating units on top of the convolutional layers at each position are also independent from each other. Therefore, our model is more suitable to parallel computing. Moreover, our model is equipped with two kinds of effective filtering mechanisms: the gating units on top of the convolutional layers and the max pooling layer, both of which can accurately generate and select aspect-related sentiment features.

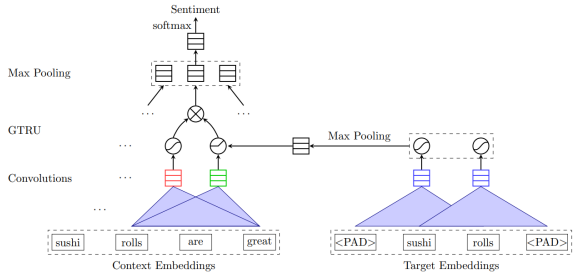


Figure 1: Illustration of GatedCNN for ABSA task.

The proposed Gated Tanh-ReLU Units control the path through which the sentiment information flows towards the pooling layer. The gating mechanisms have proven to be effective in LSTM. In aspect based sentiment analysis, it is very common that different aspects with different sentiments appear in one sentence.

Setup We let our model trained through 10 epochs, with learning rate of 10^{-4} batch size of 32. We use Adamgrad algorithmn to optimize parameters. For word embedding we use pretrained word-level PhoW2V [6] 300 dimensions for Vietnamese words. The model is trained by minimizing the cross-entropy loss between the ground-truth y and the predicted value $\hat{y}$ for all data samples.

3.3 RoBERTa

We used a method of constructing an auxiliary sentence and transform ABSA into a sentence-pair classification task [5] for the Restaurant domain using PhoBERT [3] as Pre-trained language models for Vietnamese. We fine-tuned the pre-trained model from RoBERTa [4] and achieve new state-of-the-art results on Vietnamese ABSA task.

Data processing After preprosccessing based on section 4.1.2, we formated input data to this model. Firstly, We turn aspect sentences into auxiliary sentences. Since we transform ABSA task to the Natural Languge Processing (the task of determining whether a "hypothesis" is true (entailment), false (contradiction), or undetermined (neutral) given a "premise"), we translate aspect terms to Vietnamese, eg. translating "RESTAURANT#PRICES" to "Giá nhà hàng". This helps the model learn more accurately the relationship between two sentences. After that, We label each pair of sentences, each main sentence is paired with each auxiliary sentence (each aspect). If the sentence does not contain that aspect, the label of that pair is "none". If it contains, we label the sentence according to the polarity with that aspect, namely "positive", "negative", "neutral".

PhoBERT pretrained model We use PhoBERT pretrained model - PretrainedlanguagemodelsforVietnamese [3]. PhoBERT pre-training approach is based

on RoBERTa [4] which optimizes the BERT pre-training procedure for more robust performance. That is the reason we choose RoBERTa to fine-tuned the pretrain model for this task. We use the versions PhoBERTbase with 135M prarameters pretraining data on 20GB of Wikipedia and News texts.

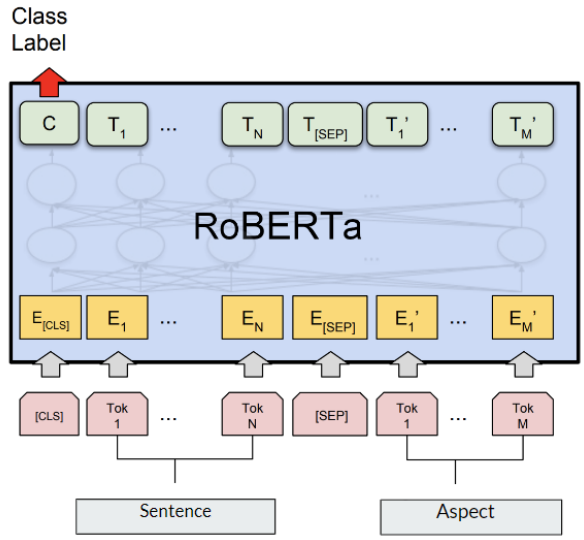


Figure 2: RoBERTa Architecture for sequence-pair classification.

RoBERTa Model We fine-tuned the pre-trained model from RoBERTa [4] (short for "Robustly Optimized BERT Approach") - a variation of the Facebook AI research team's BERT (Bidirectional Encoder Representations from Transformers) approach. Similar to BERT, RoBERTa is a transformer-based language model that processes input sequences and produces contextualized representations of words in a sentence by using self-attention. RoBERTa's architecture is nearly identical to that of BERT, however the authors added a few small design adjustments to the training process and architecture to enhance the outcomes on BERT architecture. RoBERTa is a powerful and effective language model that has made significant contributions to the field of NLP and has helped to drive progress in a wide range of applications. The architecture is shown in Figure 2.

We combine the sentence and the aspect of each sample into a compound sentence and add [CLS] and [SEP](special characters used for PhoBERT). The compound sentence,which we called sentence encoder, then had to be in the form: [CLS] sentence [SEP] the aspect [SEP].Then after that, the sentence encoder was tokenized by PhoBERT tokenizer, a extended tools for RoBERTa that Hugging Face provided We created a token type list to distinguish the sentence and the aspect in the compound sentence and an attention mask to define masking used in the RoBERTa system. These components (sentence encoder, token type, attention mask) are the necessary parameters for the pretrained PhoBERT model, which was sourced from transformers library of Hugging Face. After going through the pretrained PhoBERT- base model, we used other two feed forward layers and then a softmax layer to fine-tune and convert output dimension into the number of target labels.

Setup To optimize the parameters, we use the AdamW optimization, starting with an initial learning rate of 2e-5 and the learning rate decays linearly. The proposed models are trained for 15 epochs with a batch size of 16. The loss function used is NLL (Negative log likelihood) a loss function commonly used in multi-classes classification tasks, its meaning is to take log the probability value after softmax and add the probability value of the correct answer to the average.

3.4 BART

In this part, intend to apply pre-trained language models PLMs that are based on

sequence-to-sequence (seq2seq) architecture, for example, BARTPho, a Vietnamese encoder-decoder PLM that was designed after mBART [14]. With BARTPho, we can solve the problem by generation method, that models will generate templated outputs including information on the aspect and sentiment polarity. This idea is taken from the paper Solving Aspect Category Sentiment Analysis as a Text Generation Task [7], where the authors applied the generation method with BART on English ACSA datasets and gained promising results.

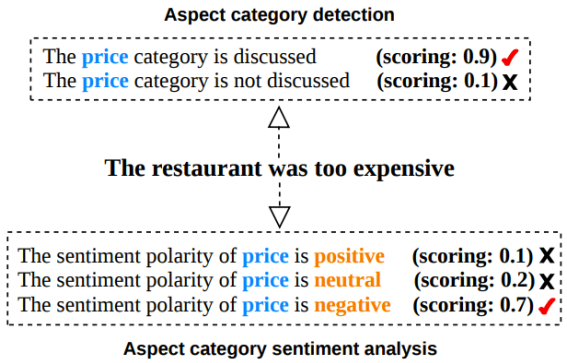


Figure 3: Example of templated outputs.

Templates As in Figure 3, the authors keep the separation of the two tasks ACD and ASC by training the Seq2seq model on templates for both tasks. For ACD, they create a sentiment template (The X category is discussed) for an existing aspect category, and a non-category template (The X category is not discussed). For ACSA, we manually create templates containing one slot for the given category (X) and another slot for the polarity-type (Y) label (The sentiment polarity of X is Y). For our model using in Vietnamese dataset, we have to translate all aspects and polarities from English to Vietnamese and construct natural Vietnamese templates ("X không/có đc nhn xét" - "giá đ ăn đc nhn xét" , "Nhn xét X Y" - "Nhn xét giá ca đ ăn tích cc."). Knowing that our Vietnamese templates are not natural due to the difficulty in translating, we still decide to use them in this part.

Inference For both ACD and ASC, we create templates for all possible circumstances of the text comment, and then use the fine-tuned pre-trained generative language model to assign a score for each template and choose the template having the highest score as the predicted value. The task ASC is only needed when the model predicts an aspect that is discussed in the comment. The score of a template $T_i = (t_1, ..., t_m)$ given the text X is calculated as below:

$$s(T_i, X) = \sum_{c=1}^{m} logP(t_c|t_{1:c-1}, X) \quad (1)$$

Training Given a text X and its label-templated sentence T, during the training stage, the text is fed into the encoder to obtain the hidden representation:

$$h^{enc} = ENCODER(X) \quad (2)$$

At the cth step of the decoder, h^{enc} and previous output tokens $t_{1:c-1}$ are then as inputs to output a representation:

$$h_c^{dec} = DECODER(h^{enc}, t_{1:c-1}) \quad (3)$$

Then we have the conditional probability of the output word at step c^{th}:

$$P(t_c|t_{1:c-1}, X) = SOFTMAX(h_c^{dec} W^{lm} + b^{lm}) \quad (4)$$

(lm: language model)

The training procedure uses the cross-entropy loss function between the label-templated sentence and the output of the decoder.

$$L = -\sum_{c=1}^{m} logP(t_c|t_{1:c-1}, X) \quad (5)$$

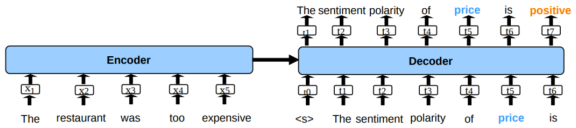


Figure 4: Example of generation method with encoder-decoder architecture.

Another strategy In addition to the two-task strategy applied in the paper, we also experiment with the ASC-only method, adding one more label ("khong co") - ("Nhan xet gia do an khong co") to inform that an aspect is not discussed.

Setup We process the texts as purposed in the previous section and then segment both the processed texts and label sentences with *pyvi* library. We use the version of BARTpho-word-base with the set-up model MBartForConditionalGeneration from *huggingface*, with a maximum tokens length of the input is 512 and a maximum generation tokens length of 32. To optimize the parameters, we use the AdamW optimization, starting with an initial learning rate of 1e-5. The proposed models of the two strategies (ACD-ACS and ACS only) are trained for 10 epochs with a batch size of 10.

4 Experiments

4.1 Datasets and Implementation

4.1.1 Metrics and Evaluation

For the evaluation, we use the F1 score as the metric calculated for all methods. Let A be the set of predicted tuples (entity#attribute, polarity), and B be the set of annotated tuples, the precision, recall and then the F1 score can be computed as follows:

$$precision = \frac{|A \cap B|}{|A|} \tag{6}$$

$$recall = \frac{|A \cap B|}{|B|} \tag{7}$$

$$F1 = \frac{2 \times precision \times recall}{precision + recall} \tag{8}$$

4.1.2 Datasets

Dataset	#Reviews	#Aspects	Avg.Len
Train	2961	9034	54
Dev	1290	3408	50
Test	500	2419	163

Table 1: The statistics for samples in the restaurant domain.

VLSP 2018 - VABSA The dataset that we use for the Vietnamese Aspect-based Sentiment Analysis (VABSA) is from VLSP 2018 - Evaluation Campaign tasks, including three disjoint parts for training (i.e., train), validation (i.e., dev) and testing (i.e., test). It consists of sets of customer reviews about services of an organization (i.e., entity) which is either restaurant or hotel, and the goal is to identify the entity for that the review is and the opinion of its. In the context, its authors express this goal as a set of tuples aspect, polarity in which the former is under the form of a tuple (entity, attribute) and the latter is the opinion (i.e., positive, negative, neutral). Take a review as the example: "Phuc vu, view đẹp, vi tri." ("Good service, sightseeing, convenient location."), and the label for this example is "{SERVICE#GENERAL, positive}, {HOTEL#GENERAL, positive}, {LOCATION#GENERAL, positive}".

All the samples are document-level reviews composing of many sentences with different

lengths and text styles. The dataset, as mentioned, contains two domains which are restaurant and hotel; we will try to resolve the former. The statistics of the dataset for restaurant domain is shown in the table 1.

In restaurant domain, there are 12 different aspects (i.e., pairs of entity-attribute) and 3 polarities as mentioned. The problem is, therefore, a multitask classification problem with the former being multi-label and the latter multi-class one. In particular, entities can be RESTAURANT (in general), AMBIENCE, LOCATION, FOOD, DRINKS, or SERVICE; attributes can be GENERAL, QUALITY, PRICE, STYLE & OPTIONS, or MISCELLANEOUS. . The possible combinations of these entities and attributes are given in [12]. Moreover, The amount of data on each aspect in test data is presented in figure 5.

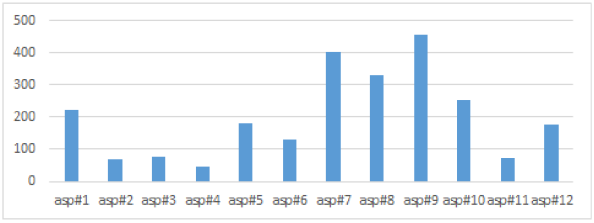


Figure 5: The amount of data for each aspect in Restaurant dataset. The aliases are specified in the dataset note [12].

Preprocessing For machine to learn the languages efficiently, text processing and text vectorization are the must-be steps. Since the emerge of modern architectures (e.g., transformer), the vectorization is now included in the models as a whole. Therefore, we do only perform text cleaning and tokenization for the whole methods and leave the other preprocessing schemes optional for each method separately. As the data is about reviewing of restaurants, the writing is not paid attention and many irregular wordings appear across reviews (e.g., "khong / no" can be written as "ko", "k", "khg", etc.). Therefore, we propose a general pipeline to standardize and tokenize the text, as shown in the figure 6.

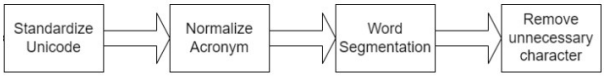


Figure 6: The general text processing pipeline that is used for all models.

First, we perform the charset standardization (i.e., to UTF-8) and then normalize common Vietnamese abbreviations for the dataset, as follows:

- Price,hashtag and website normalization: there are different ways that prices are perform in the training set such as "50k", "3tr", we replace them with the single word "giá". Different hashtags #abc are also replaced by "tag". The same with websites, any link or website reference is replaced with the word "website".

- Emoji normalization: Since emojis play an important role in extracting sentiment, so we replace each emoji by one word that shows its status. For example, emojis that show happy or exiting emotions are replaced by "tích cuc" and emojis that show negative feeling are replace by the word "tiêu cuc".

- Elongate word normalization: since the data includes comments crawled from the Internet so there are many elongate words such as "thíchhhhh" , "vuiiiii". We turn them into their correct forms by removing unnecessary character.

- Teencode processing: this system of words is complicated there hasn't existed a tool that can handle this teencode problem so the only solution is to replace them directly. We create our own dictionary that

process each teencode and return its regular form that our model can understand.

Since an vietnamese word can be made from up to two or more words working together (i.e., compound words), it is essential to tokenize such phrases before starting any further; here, we utilize VNCoreNLP toolkit [1]. At last, we remove special characters (i.e., characters out of vietnamese alphabets; e.g., @, <, =, $) and extra whitespaces. We expect that this preprocessing stage will be sufficient to produce a quality dataset of text reviews that leverages the learning ability of the models.

4.2 Results and Discussion

In this section, we present our experiments and results for the two tasks - aspect detection and aspect-polarity in the restaurant reviews. In particular, the models are evaluated for whether they tag enough aspects (*enitity#attribute*) correctly and then for both tags and their corresponding polarity (*enitity#attribute*, *polarity*). We show the results for the models used in this work in table 2 and table 3.

Methods	Precision	Recall	F1
LDA + RF	0.80	0.41	0.53
LDA + SVM	0.76	0.51	0.61
Gated CNN	0.74	0.70	0.74
BART(ACD/C)	0.83	0.85	0.83
BART(ASC)	0.84	0.83	0.82
RoBERTa	**0.86**	**0.86**	**0.85**

Table 2: The performance of different methods on test dataset of aspect detection task.

The table 2 describes the precision, recall and F1-score of the methods measured in test set. As shown, RoBERTa achieves the highest performance on both tasks over the three different metrics. In the aspect detection task,

Methods	Precision	Recall	F1
LDA + RF	0.65	0.34	0.43
LDA + SVM	0.74	0.46	0.55
Gated CNN	0.70	0.65	0.67
BART(ACD/C)	0.70	0.72	0.70
BART(ASC)	0.72	0.71	0.71
RoBERTa	**0.81**	**0.81**	**0.80**

Table 3: The performance of different methods on test dataset of aspect-polarity task.

the two strategies of BARTpho follow closely our best model with only 0.2-0.3 difference on each metric. However, RoBERTa shows its robustness on the harder task (table 3) which demands to detect both the aspects and polarities as a whole. To summarize, RoBERTa beats other methods significantly with 0.8 F1, which nearly doubles the performance of classical models, and by far is our best method for aspect-based sentiment analysis overall. For BARTpho, the two-task ACD-ASC approach seems to have a little higher recall scores compared with only ASC; however, both show disappointing performances when detecting sentiment polarities. A possible reason may come from the unnatural translation of templates, aspects and polarities as in the original paper [7] the authors state that the quality of the templates does affect the model's performance.

Comparison to other researches We aim to extend our results, comparing the performance of our best model to other proposals that work on the same domain. In particular, the methods mentioned in Hoang-Quang Dang's work [8] are brought out to assess the results of RoBERTa.

As shown on the table 4 and 5, our best method is still on the peak as it outperforms other proposals significantly, especially in the aspect-polarity task. BERT-based hierarchical model is able to approach the performance of

Methods	Precision	Recall	F1
VLSP Submit.	0.79	0.76	0.77
BiLSTM CNN	0.82	0.76	0.80
BERT (hier.)	-	-	0.84
Multi-task	0.81	0.86	0.83
Multi-branch	0.81	0.87	0.84
RoBERTa	**0.86**	**0.86**	**0.85**

Table 4: The performance of other proposals [8] vs. ours in aspect detection task.

Methods	Precision	Recall	F1
VLSP Submit.	0.62	0.60	0.61
BiLSTM CNN	0.67	0.63	0.65
BERT (hier.)	-	-	0.71
Multi-task	0.70	0.73	0.72
Multi-branch	0.69	0.74	0.71
RoBERTa	**0.81**	**0.81**	**0.80**

Table 5: The performance of other proposals [8] vs. ours in aspect-polarity task.

RoBERTa in the first task. However, the same situation happens when it comes to the harder task when RoBERTa shows its dominance.

5 Conclusion

In this article, we have solved the problem of Aspect-based Sentiment analysis on the VSLP 2018 ASBA dataset. We have implemented modern models as BARTPho, RoBERTa and traditional models as LDA + SVM, LDA + RF and Gated CNN. In which, the RoBERTa model gives the best result on Restaurant domains when applying the preprocessing and using PhoBERT pretrained model. Specifically, the RoBERTa model achieves state-of-the-art (SOTA) results for the Restaurant domain when compared to other results on benchmark. In addition, through the experimental model, we find that the model for Natural Language Inference is greatly effective on ABSA task.

In the future, we want to try other variants of RoBERTa model for this problem such as TFRobertaForSequenceClassification, FlaxRobertaForSequenceClassification to test the effectiveness. Moreover, we will make some improvements for out SOTA model such as using Probabilistic Linear Discriminant Analysis (PLDA) for classification.

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